Analyzing Toronto Apartment Rental Prices

Austin Hart Olivia Jiang Woobin Kim Raina Tian

August 2, 2026

Project Task Assignments

Raina (11946720) wrote most of the introduction and data description.

Olivia (82785783) (group leader) did most of the exploratory data analysis section.

Austin (45726379) did most of the modeling section.

Woobin (39017496) wrote most of the conclusion.

All members contributed to editing and proofreading the project as a whole.

Introduction

In high-density metropolitan areas like Toronto, assessing how specific layout configurations influence rental pricing provides valuable insight into market structure and housing affordability. This study examines the relationship between apartment layout characteristics and monthly rent across the city. Specifically, we use multiple linear regression to evaluate how the number of bedrooms, bathrooms, and any interaction effects predict continuous rental prices. To test predictive validity further, we apply binary logistic regression using the same structural features to determine whether a listing classifies as a high-value property. By comparing main-effects and interaction models across both regression frameworks, we assess the overall explanatory power of layout features in Toronto's 2018 housing market.

Data Description

The dataset for this study was retrieved from a public Kaggle repository titled Toronto Apartment Rental Price, compiled by independent creator Raja CSP. According to the dataset documentation, the listings were aggregated and scraped from local rental websites active across Toronto in 2018.

The study utilizes Version 3 of the dataset, stored as a single 94.97 KB comma-separated values file. The raw file contains 1,124 observational rows, where each observation is a unique apartment listing with 7 attributes: its monthly rental price, address, latitude, longitude, and amount of bedrooms, bathrooms, and dens. There are no missing values in the raw data.

Exploratory Data Analysis

During data cleaning, we removed 308 duplicates which likely occurred because the data came from multiple sources. We also removed 9 rows where the price was either below $100/month or above $6,000/month (namely, $65, $99, $99, $6,000, $6,500, $8,000, $9,750, $36,900, and $535,000) because these outliers may skew our results. The sample size is now `n = 807`.

Before modeling, we want to find out more about the data.

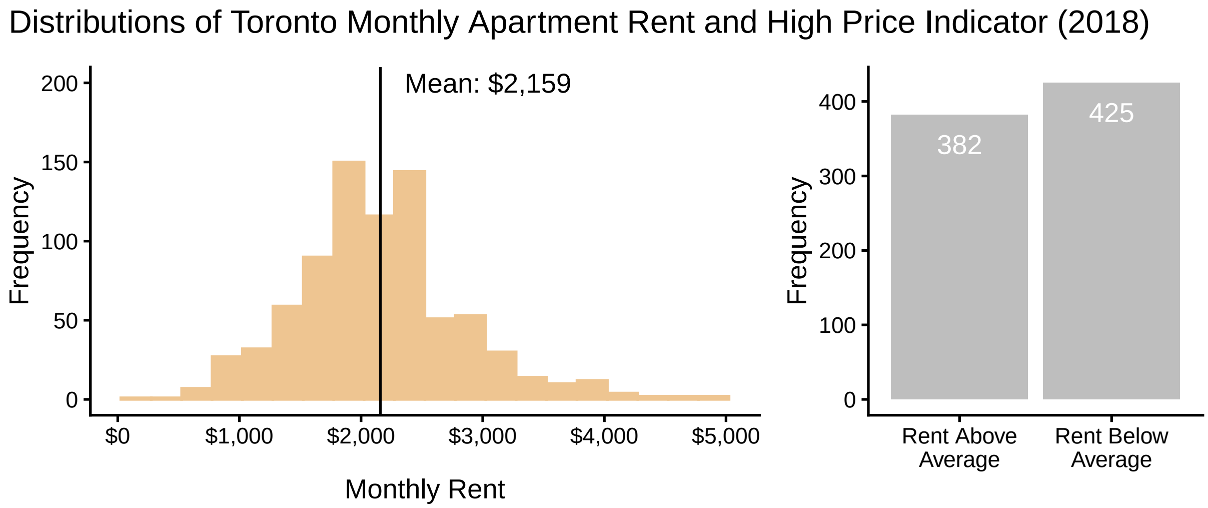
We first examine sample's summary statistics of each individual numerical variable.

Variable	Mean	Standard Deviation	Minimum	Maximum
bathroom	1.23	0.43	1.00	3.00
bedroom	1.36	0.56	1.00	3.00
den	0.15	0.36	0.00	1.00
lat	43.71	0.69	42.99	56.13
long	-79.50	1.85	-114.08	-73.58
price	2159.05	684.27	150.00	4900.00

The amount of bathrooms ranges from 1 to 3, the amount of bedrooms ranges from 1 to 3, and the amount of dens ranges from 0 to 1. The average amount of bedrooms is slightly higher than the average amount of bathrooms, and most apartment's don't have any dens, shown by its mean of 0.15. This makes sense - an apartment should have a bedroom and bathroom whereas a den is optional. Also, latitude and longitude fall in the valid range of [-90, 90] and [-180, 180] respectively (though we won't use these variables in modeling because geographic effects on a target variable are probably not as linear or easily linearly interpretable). So there's nothing unusual about these four variables.

As for `price`, its average value is $2,159/month with a standard deviation of $684/month. Its minimum is $150/month and its maximum is $4,900/month. While these minimum and maximum values are -2.59 and 4.01 standard deviations above and below the mean (respectively), we feel that they are reasonable enough to keep.

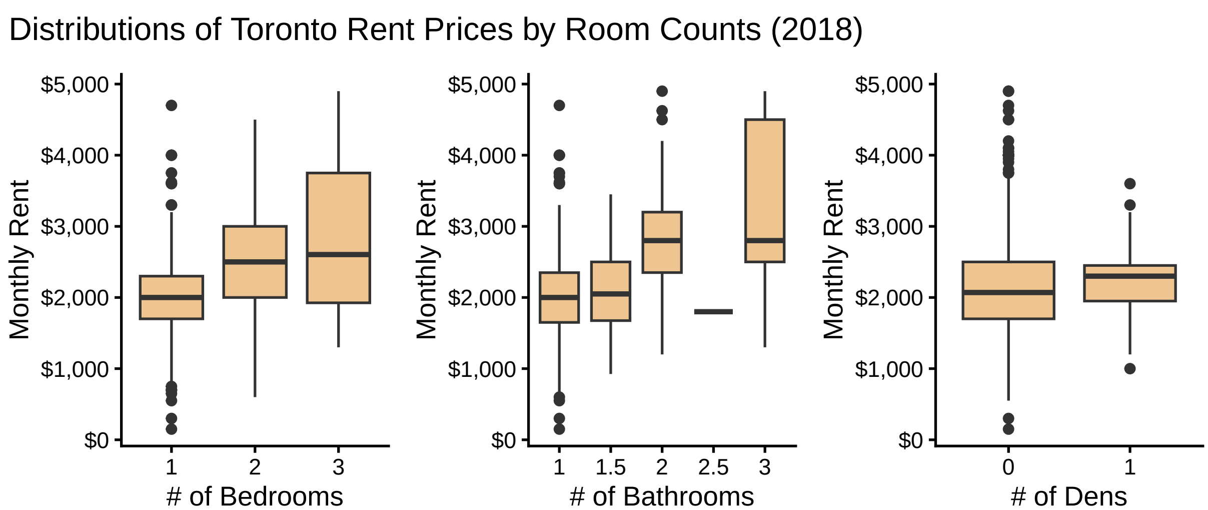
We now plot a histogram to visualize the distribution of price and a bar chart showing the class balance of the binary variable `high_price` we created, which indicates if the price is above the average price of $2,159/month or not.



On the left, we see that `price` is approximately symmetric and unimodal around its mean. On the right, we see that the two `high_price` groups are approximately balanced; this means that logistic regression won't run into the issue of unbalanced response groups, which could bias modeling results.

Next, we examine possible relationships *between* variables.

Below are boxplots of `price` split by `bedroom`, `bathroom`, and `den` counts to see how `price`'s distribution varies by each of these apartment characteristics.



Looking independently at `bedroom`, `price`'s distribution slightly shifts upwards by a few hundred dollars for each additional bedroom. For `bathroom`, the distribution also moves upwards with the biggest drop from 2 to 2.5 bathrooms and another upwards shift in the distribution from 2.5 to 3 bathrooms. For `den`, the distribution is roughly equal for both groups with a difference-between-means of only $87/month. As well, within each group of boxplots, the distributions overlap with each other, so these predictors may not produce strong modeling results. Nonetheless, in terms of potential modeling power, `bathroom` may be better than `bedroom` and it may be better not to use `den`.

We're also curious about the correlations between the variables so we plot the upper triangle of a (Pearson) correlation matrix. Pearson correlation (r) falls in [-1, 1]. If two variables are positively correlated, it means that as one gets larger, the other also tends to get larger. If two variables are negatively correlated, it means that as one gets larger, the other tends to get smaller.

Looking at the rightmost column, the variables with the strongest linear correlation with `price` are `bathroom` (r = 0.47) then `bedroom` (r = 0.39), though these are only moderately strong. In addition to the boxplots' evidence, the matrix further confirms that `den` is weakly correlated with `price` at r = 0.05. And even though `bedroom` and `bathroom` are moderately correlated at r = 0.63, putting models at risk of multicollinearity, both variables measure different characteristics of an apartment, so we may keep both during modeling.

In addition, here are frequency tables between `high_price` and `bedroom` and `bathroom`, excluding `den` because we've already seen that it would be a weak predictor.

# of Bedrooms	Rent Below Average	Rent Above Average	Total
1	62% (340)	38% (208)	100% (548)
2	32% (72)	68% (153)	100% (225)
3	38% (13)	62% (21)	100% (34)
Total	53% (425)	47% (382)	100% (807)

# of Bathrooms	Rent Below Average	Rent Above Average	Total
1	62% (381)	38% (232)	100% (613)
1.5	52% (12)	48% (11)	100% (23)
2	18% (30)	82% (135)	100% (165)
2.5	100% (1)	0% (0)	100% (1)
3	20% (1)	80% (4)	100% (5)
Total	53% (425)	47% (382)	100% (807)

From these tables, we see that of the data is on 1-bedroom or 1-bathroom units with the least data on units with 1.5, 2.5, or 3 bathrooms. Also, expectedly, most (62%) of units with a price below average have 1 bedroom and 1 bathroom.

The main findings of this exploratory analysis are:

1. There is no class imbalance in `high_price` (for the purpose of logistic regression) because 53% vs. 47% of the observations have a `price` above vs. below average. Class imbalance, such as 90% vs. 5%, could bias a model's intercept and produce a poor model that maximizes accuracy by simply predicting the majority class.
2. In terms of predictive ability, `den` likely has the weakest linear relationship with `price`, as shown by the very similar boxplots and Pearson correlation of 0.05, and we exclude `latitude` and `longitude` because geographic effects on a target variable are probably not as linear or straightforward as these coordinates. This leaves `bedroom` and `bathroom` as the covariates remaining.

Models

Now that we have explored the data, we can consider models that can model rental price based on other variables. In the previous section, we narrowed down the variables we can use as predictors to the number of bedrooms and the number of bathrooms. To make use of these predictors, we chose two model types: a multiple linear regression on `price` because it is continuous and a multiple logistic regression on `high_price` because it is binary.

Multiple Linear Regression

A multiple linear regression model models a continuous response variable in terms of the predictors as a straight line with an intercept and at least one slope.

When making this model type, we often consider whether interaction between covariates is statistically significant. If it is, then the slope of the prediction line differs depending on the value of the covariates, otherwise the slope is constant. Statistical significance can be checked with hypothesis testing by creating a model with interaction and checking its statistics.

Results with interactions:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1496.02	210.86	7.09	0.00	1082.12	1909.93
bedroom	0.95	125.45	0.01	0.99	-245.30	247.21
bathroom	318.44	180.91	1.76	0.08	-36.68	673.56
bedroom:bathroom	146.84	90.94	1.61	0.11	-31.67	325.35

Results without interactions:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1172.76	66.25	17.70	0.0000000	1042.72	1302.80
bedroom	187.90	48.35	3.89	0.0001101	93.00	282.80
bathroom	592.21	63.16	9.38	0.0000000	468.23	716.19

In the first table, the interaction term (`bedroom:bathroom`) has a p-value of 0.11 in excess of our pre-decided level of significance of 0.05. Thus, we cannot confidently identify an interaction between the predictors and choose to proceed with the model without interaction, which has small p-values for all three coefficients.

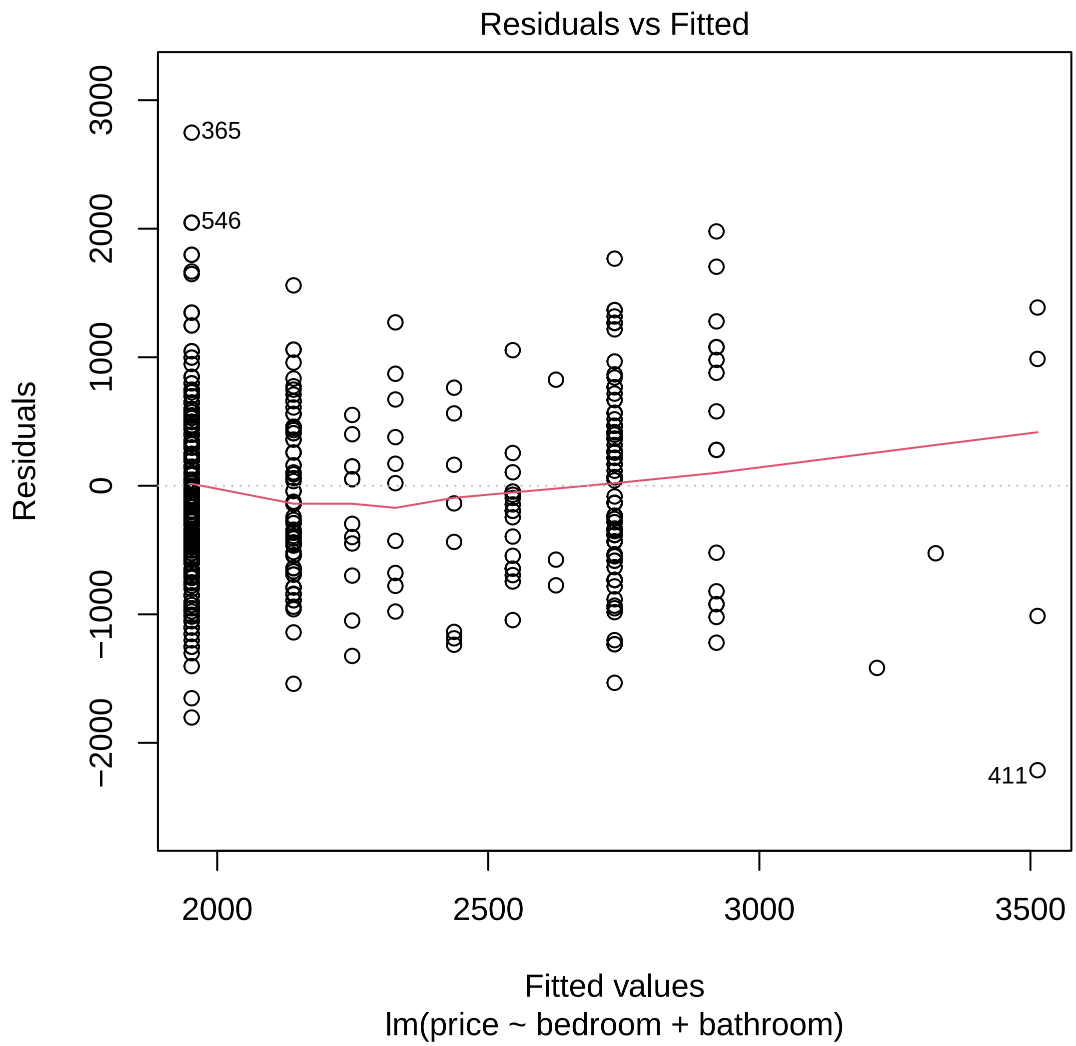
The `estimate` column in the second table shows a prediction line of the form:

$$\text{price} = \$1172.76 + \$187.90 \cdot \text{bedroom} + \$592.21 \cdot \text{bathroom}$$

The intercept value has no obvious interpretation because a place with no bedrooms or bathrooms isn't very realistic; however, the intercept is like a necessary starting point for the prediction so that the slopes stay unbiased. Both slopes are statistically significant at the 0.05 level and `bathroom` has a smaller p-value than `bedroom`, consistent with the EDA results that `bathroom` may be a slightly better predictor. The slope values show that each additional bedroom in a property is associated with a \$187.90 increase in monthly rent when the number of bathrooms is fixed at any value, and that an additional bathroom is associated with a \$592.21 increase in rent when the number of bedrooms is fixed at any value. For example, the model predicts that a one-bed, one-bath condo costs \$1172.76 + \$187.90 + \$592.21 = \$1952.87/month.

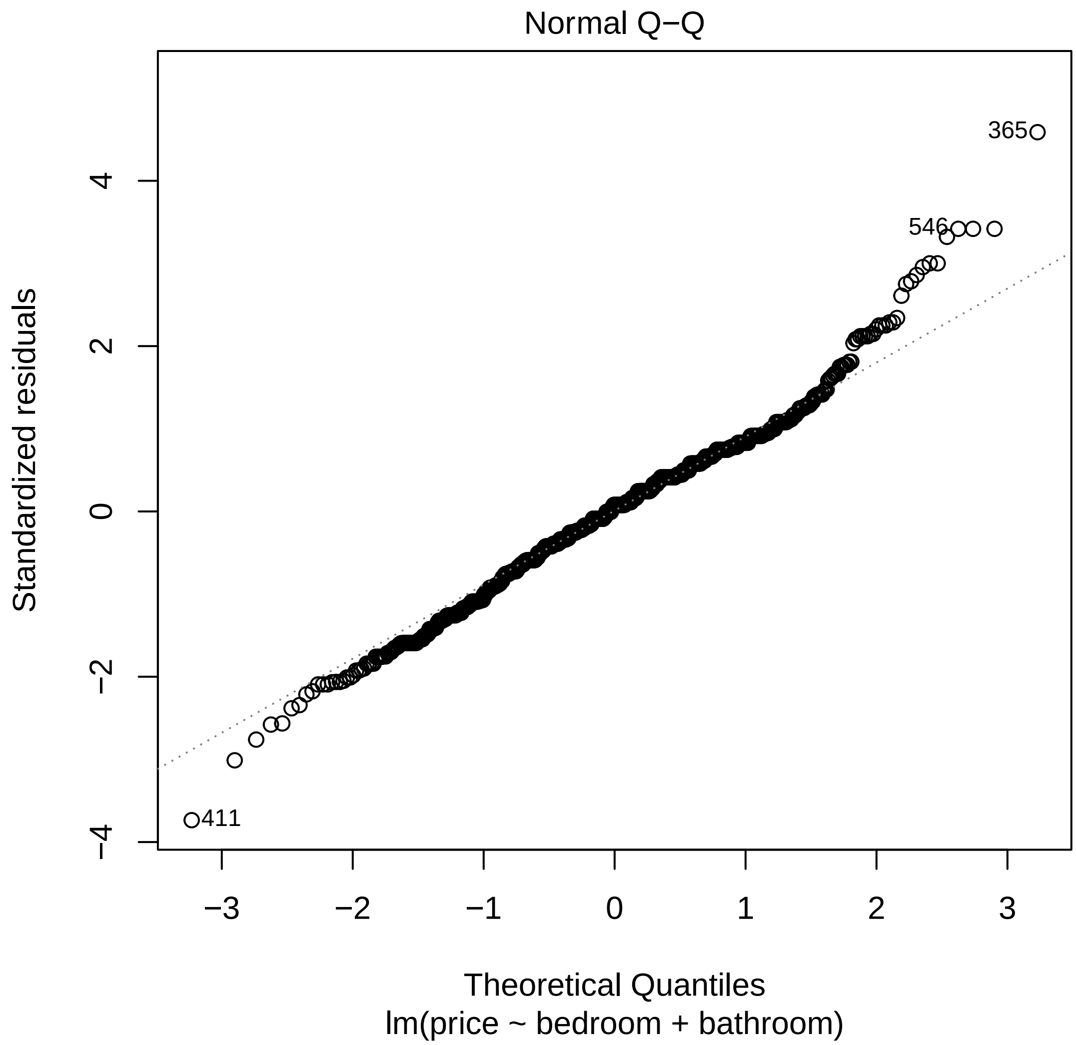
To check if the fitted model is appropriate, we create visual diagnostics in the form of a residual plot and a Q-Q plot. These essentially check if the assumptions for linear regression to be valid are fulfilled.

A residual plot plots the residuals/errors (the difference between the actual and fitted values) against the fitted values to check for the errors' linearity (i.e., a mean of 0) and constant variance. Errors should be distributed normally and evenly around the e = 0 horizontal line.



The error appear to have a mean of 0 because the points appear evenly around both sides of e = 0, but the variance is larger at certain areas (like when the fitted values are around 2000 and 3000), so the variance is not perfectly consistent. Applying transformations to the response variable, such as a log transformation and a square root transformation, were unsuccessful in creating a more favourable distribution.

A Q-Q plot is another way to check the errors' distribution. If the error is normally distributed, then a Q-Q plot shows the data points closely distributed along a straight, sloped line wherein x = y.



This Q-Q plot shows the data following this line for the most part, but the points diverge at the beginning and the end, implying that the model has more extreme data than can be expected under a true normal distribution (Ford, 2015). Applying transformations to the response variable, such as a log transformation and a square root transformation, were unsuccessful in creating a more favourable distribution.

While our diagnostics reveal that our data may not be perfectly suited for a linear analysis, this is not uncommon for observed data. It is our belief that our model can still be useful for predicting.

Multiple Logistic Regression

A multiple logistic regression model is used to make predictions for a binary response variable so we made `price` binary by splitting it into two categories: above the mean price or not. So our model will predict whether a property has above or below average rent rather than a specific value.

Once again, we begin by checking for interaction between predictors.

Results with interactions:

Log-Odds of `high_price`

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.30	0.17	-1.82	0.07	-0.62	0.02
bedroom	0.20	0.10	2.05	0.04	0.01	0.39
bathroom	0.58	0.14	4.13	0.00	0.31	0.86
bedroom:bathroom	-0.12	0.07	-1.70	0.09	-0.26	0.02

Like our linear model, we cannot conclude that there is significant interaction between the predictors at a 0.05 significance level because the p-value of `bedroom:bathroom` is 0.09.

Results without interactions:

Log-Odds of `high_price`

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.09	0.26	-9.33	0.0000000	0.05	0.15
bedroom	1.24	0.17	1.27	0.2041911	0.89	1.73
bathroom	5.25	0.24	6.81	0.0000000	3.29	8.55

Odds of `high_price`

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.09	0.26	-9.33	0.0000000	0.05	0.15
bedroom	1.24	0.17	1.27	0.2041911	0.89	1.73
bathroom	5.25	0.24	6.81	0.0000000	3.29	8.55

The `estimate` column in the table of the log-odds model without interactions shows a prediction line of the form:

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = -2.41 + 0.22 \cdot \text{bedroom} + 1.66 \cdot \text{bathroom}$$

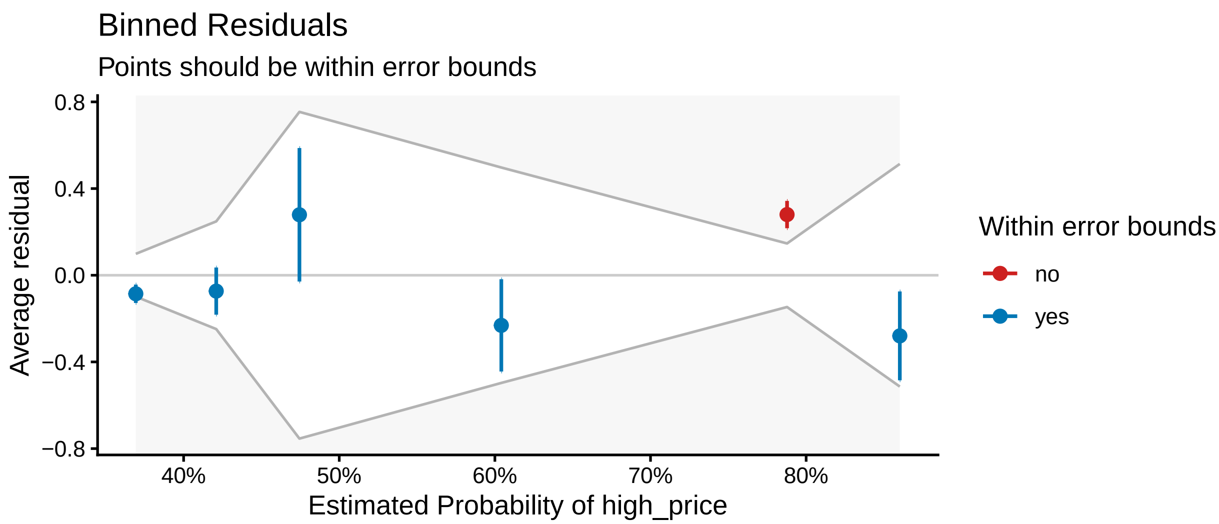
where $\hat{p}$ is the expected probability of a property having high rent and $\frac{\hat{p}}{1-\hat{p}}$ is its odds.

This means that an additional bedroom increases the expected *log-odds* of a property having high rent by 0.05 and increases the expected *odds* of a property having high rent by a factor of $e^{0.22} = 1.24$ or 24%. However, the p-value of 0.20 for this estimate implies that this conclusion is not statistically significant at a 0.05 significance level. Similarly, an additional bathroom increases the odds of a property having high rent by $e^{1.66} = 5.25$ or 425%, which is statistically significant.

Again, we do a few model diagnostics.

One of the first things to check for is overdispersion, which occurs when our response distribution deviates from the assumption that the distribution of the fitted values is consistent with a Bernoulli distribution: the model has a dispersion parameter of 1.02, implying an insignificant amount of overdispersion.

As an additional assessment, we use a binned residuals plot, where most data points should fall within the generated error bounds for the model to be considered appropriate (Lüdecke et al., n.d.).



Each dot is the average residual for one bin of predicted probabilities and the vertical lines show the uncertainty around that average residual. The gray lines show the acceptable range for residuals. Since 1 of 6 bins fall outside the range, at the estimated probabilities exceeding 90%, the plot shows that the model fits decently well and that higher predicted probabilities should be more carefully interpreted.

Conclusion

Both the exploratory data analysis and the multiple linear regression analysis indicate that both the number of bedrooms and bathrooms have a positive relationship with monthly rent. Specifically, holding other variables constant, an additional bedroom increases predicted monthly rent by $187.90 ($p < 0.001$), while an additional bathroom increases rent by $592.21 ($p < 0.001$). The multiple logistic regression analysis indicates that bathrooms have a statistically significant positive effect on the odds of a property having an above-average rent ($p < 0.001$), increasing the odds by 425% per bathroom, while the effect of bedrooms is not statistically significant at the standard 0.05 level ($p = 0.21$).

However, as seen from the residual plot and the Q-Q plot, the assumptions of homoscedasticity and the normality of residuals for the linear model are not completely valid. These violations affect prediction confidence intervals and the p-values such that the current linear model results cannot be considered fully reliable. Another key assumption of the analysis is that the sample is a simple random sample and that each observation is independent of the others. Consequently, findings regarding property characteristics should be interpreted with caution given potential omitted variable bias and the uncertainty of these assumptions.

To improve model reliability in future research, we suggest some adjustments. For example, re-introducing location by creating neighbourhood clusters may help reduce unexplained variance and standard errors, thereby addressing potential omitted variable bias. As well, finding continuous structural variables like square footage and property age alongside discrete room counts may help normalize the residual distribution and provide more depth and detail to any resulting models.

References

Ford, C. (2016, August 26). *Understanding QQ plots.* University of Virginia Library. https://library.virginia.edu/data/articles/understanding-q-q-plots

Lüdecke, D., Makowski, D., Ben-Shachar, M. S., Patil, I., Waggoner, P., Weirnik, B. M., & Thériault, R. (n.d.). *Binned residuals for binomial logistic regression.* Easystats. Retrieved July 31, 2026, from https://easystats.github.io/performance/reference/binned_residuals.html.

Appendix

```
# Loading packages
library(tidyverse)
library(knitr)
library(scales)
library(patchwork)
library(corrplot)
library(broom)
library(janitor)
library(performance)
library(see)

# Reading the data
data <- read_csv("Toronto_apartment_rentals_2018.csv")

# Seeing how many duplicate rows there are
sum(duplicated(data)) # 308

# Viewing duplicate rows
data[duplicated(data) | duplicated(data, fromLast = TRUE), ]

# Cleaning the data
data <- data %>%
  # Removing duplicate rows
  distinct() %>%
  # Removing unnecessary columns
  select(-c(Address)) %>%
  # Making the column names lowercase
  rename_with(tolower) %>%
  # Converting price to a number
  mutate(price = parse_number(price)) %>%
  # Removing outliers
  filter(price > 100 & price < 6000)

# Seeing if there are any NAs
colSums(is.na(data)) # 0

# Adding a binary response variable (whether price is more than its mean)
data <- data %>%
  mutate(high_price = if_else(price > mean(price), 1, 0))

# Viewing summary stats
data %>%
  select(-high_price) %>%
  # Reshaping data into a long format
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value") %>%
  # Grouping by the variable
  group_by(Variable) %>%
  # Calculating the stats
  summarise(Mean = round(mean(Value, na.rm = TRUE), 2),
            "Standard Deviation" = round(sd(Value, na.rm = TRUE), 2),
            Minimum = round(min(Value, na.rm = TRUE), 2),
```

```r
            Maximum = round(max(Value, na.rm = TRUE), 2)) %>%
  kable()

# Visualizing response variables

hist_1 <- data %>%
  ggplot(aes(x = price)) +
  geom_histogram(fill = "burlywood2", color = "burlywood2", bins = 20) +
  labs(x = "Monthly Rent",
       y = "Frequency") +
  geom_vline(aes(xintercept = mean(price)),
             color = "black", linewidth = 0.5) +
  annotate("text",
           x = 2159 + 200,
           y = 200,
           label = "Mean: $2,159",
           color = "black",
           hjust = 0) +
  scale_x_continuous(labels = label_dollar()) +
  theme_classic()

hist_2 <- data %>%
  mutate(high_price = ifelse(high_price == 0,
                             "Rent Below\nAverage",
                             "Rent Above\nAverage")) %>%
  ggplot(aes(x = as.factor(high_price))) +
  geom_bar(fill = "grey") +
  labs(x = "", y = "Frequency") +
  geom_text(stat = "count", aes(label = after_stat(count)),
            vjust = 2, color = "white") +
  theme_classic()

hist_1 + hist_2 + plot_layout(widths = c(2, 1)) + plot_annotation(
  title = "Distributions of Toronto Monthly Apartment Rent and High Price Indicator (2018)"
  )

# Making boxplots of price by bed/bath/den count

boxplot_1 <- data %>%
  mutate(bedroom = as.factor(bedroom)) %>%
  ggplot(aes(x = bedroom, y = price)) +
  geom_boxplot(fill = "burlywood2") +
  labs(y = "Monthly Rent", x = "# of Bedrooms") +
  scale_y_continuous(labels = label_dollar()) +
  theme_classic()

boxplot_2 <- data %>%
  mutate(bathroom = as.factor(bathroom)) %>%
  ggplot(aes(x = bathroom, y = price)) +
  geom_boxplot(fill = "burlywood2") +
  labs(y = "Monthly Rent", x = "# of Bathrooms") +
  scale_y_continuous(labels = label_dollar()) +
  theme_classic()
```

```
boxplot_3 <- data %>%
  mutate(den = as.factor(den)) %>%
  ggplot(aes(x = den, y = price)) +
  geom_boxplot(fill = "burlywood2") +
  labs(y = "Monthly Rent", x = "# of Dens") +
  scale_y_continuous(labels = label_dollar()) +
  theme_classic()

boxplot_1 + boxplot_2 + boxplot_3 + plot_annotation(
  title = "Distributions of Toronto Rent Prices by Room Counts (2018)"
  )

# Making a correlation matrix

par(font = 1)

corrplot(round(cor(select(data, -c(high_price, lat, long))), 2),
         method = "color",
         type = "upper",
         addCoef.col = "black",
         tl.col = "black",
         tl.srt = 30,
         tl.cex = 0.7,
         cl.ratio = 0.35,
         cl.length = 5,
         cl.cex = 0.7,
         number.cex = 0.7,
         col = colorRampPalette(c("#4477AA", "#77AADD", "#FFFFFF", "#EE9988", "#BB4444"))(200),
         mar = c(0, 0, 1, 0),
         diag = TRUE)

# Making frequency tables

data %>%
  tabyl(bedroom, high_price) %>%
  adorn_totals(c("row", "col")) %>%
  adorn_percentages("row") %>%
  adorn_pct_formatting(digits = 0) %>%
  adorn_ns() %>%
  kable(col.names = c("# of Bedrooms", "Rent Below Average", "Rent Above Average", "Total"))

data %>%
  tabyl(bathroom, high_price) %>%
  adorn_totals(c("row", "col")) %>%
  adorn_percentages("row") %>%
  adorn_pct_formatting(digits = 0) %>%
  adorn_ns() %>%
  kable(col.names = c("# of Bathrooms", "Rent Below Average", "Rent Above Average", "Total"))

# Linear multiplicative model

linear_mult <- lm(price ~ bedroom * bathroom, data = data)
```

```r
tidy(linear_mult, conf.int = TRUE, conf.level = 0.95) %>%
  mutate_if(is.numeric, round, 2) %>%
  kable()

# Linear additive model

linear_add <- lm(price ~ bedroom + bathroom, data = data)

tidy(linear_add, conf.int = TRUE, conf.level = 0.95) %>%
  mutate(across(where(is.numeric) & !p.value, ~ round(.x, digits = 2))) %>%
  kable()

# Residual plot

plot(linear_add, which = 1)

# Q-Q plot

plot(linear_add, which = 2)

# Multiplicative logistic model

logistic_mult <- glm(high_price ~ bedroom * bathroom, data = data)

tidy(logistic_mult, conf.int = TRUE, conf.level = 0.95) %>%
  mutate_if(is.numeric, round, 2) %>%
  kable()

# Additive logistic model

logistic_add <- glm(high_price ~ bedroom + bathroom, data = data)

tidy(logistic_add, conf.int = TRUE, conf.level = 0.95) %>%
  mutate(across(where(is.numeric) & !p.value, ~ round(.x, digits = 2))) %>%
  kable()

tidy(logistic_add, conf.int = TRUE, conf.level = 0.95, exponentiate = TRUE) %>%
  mutate(across(where(is.numeric) & !p.value, ~ round(.x, digits = 2))) %>%
  kable()

# Overdispersion

logistic_add_disp <- glm(high_price ~ bedroom + bathroom, data = data, family = "quasibinomial")

summary(logistic_add_disp)

# Binned residuals plot

plot(binned_residuals(logistic_add)) +
  theme_classic()
```